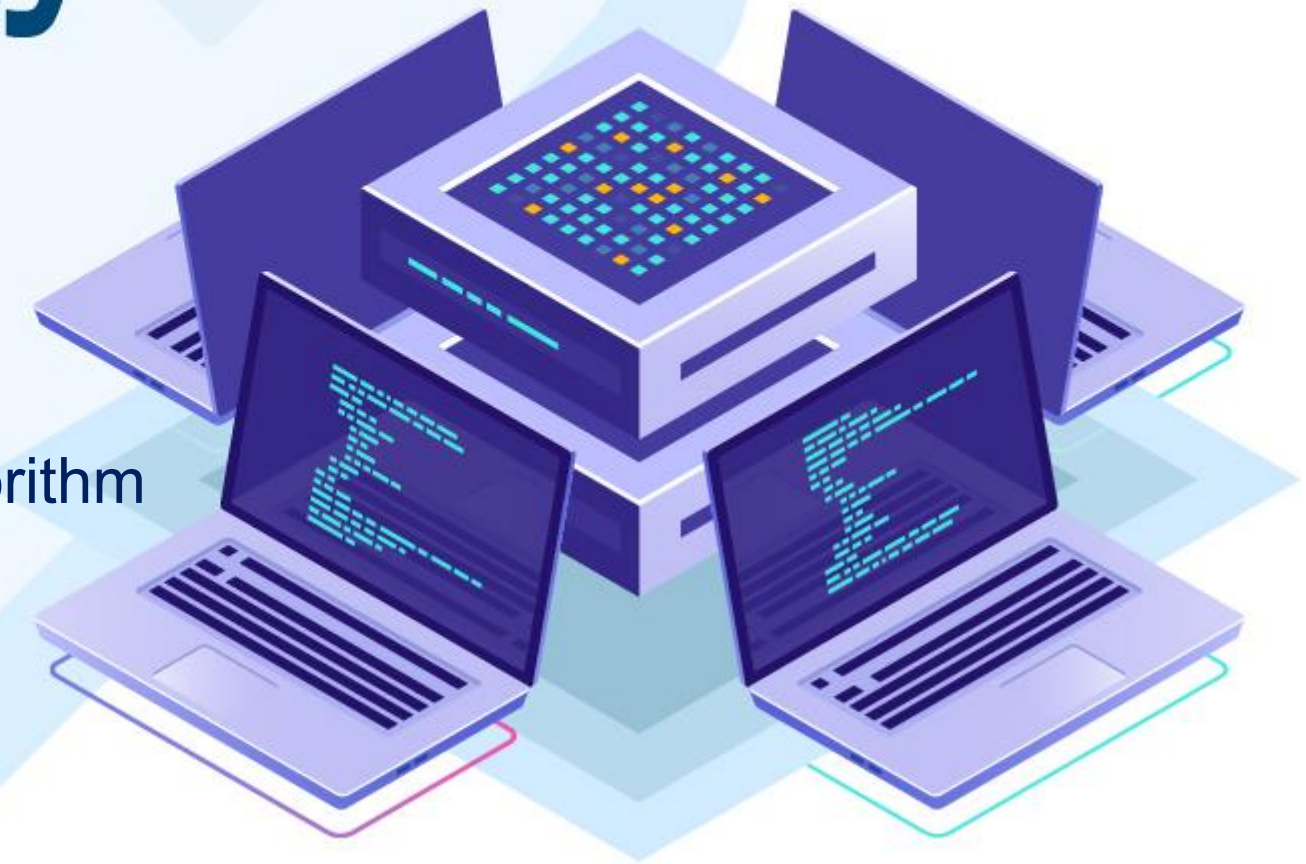
Information Technology

Year: 2026-2027

Fall Semester

- Computer Graphics
- **Dr. Yasmine Mahmoud**

Bresenham’s Line Drawing Algorithm



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الجامعة المصرية للتعلم الإلكتروني الأهلية

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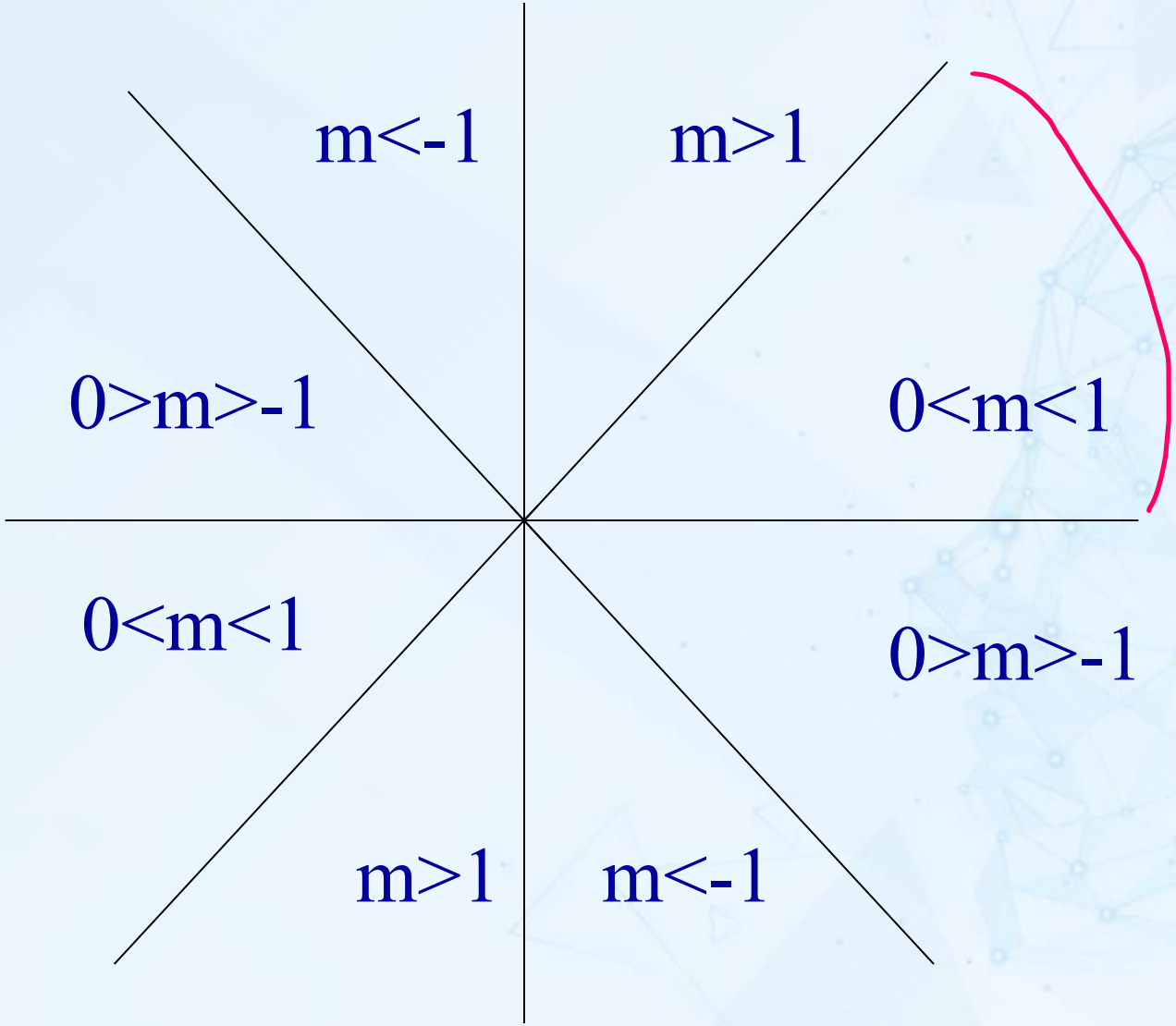
Bresenham's Line Drawing Algorithm

- The basic principle of Bresenham's line algorithm is to select the optimum raster locations to represent a straight line.
- To accomplish this algorithm always increments either x or y by one unit depending on the slope of line.

- Accurate and efficient
- Uses only incremental integer calculations

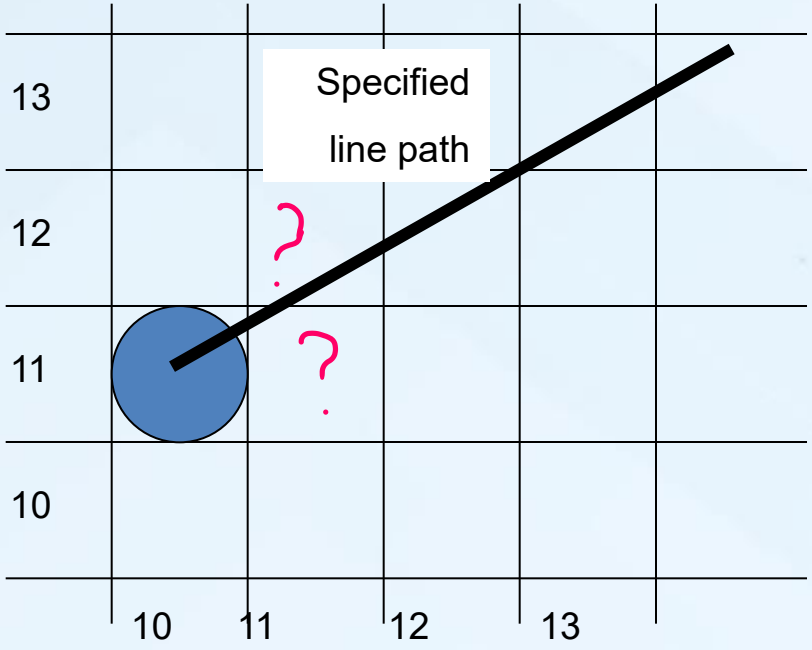
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Line-drawing algorithm should work in every octant, and special cases



The method is described for a line segment with a **positive slope less than one**

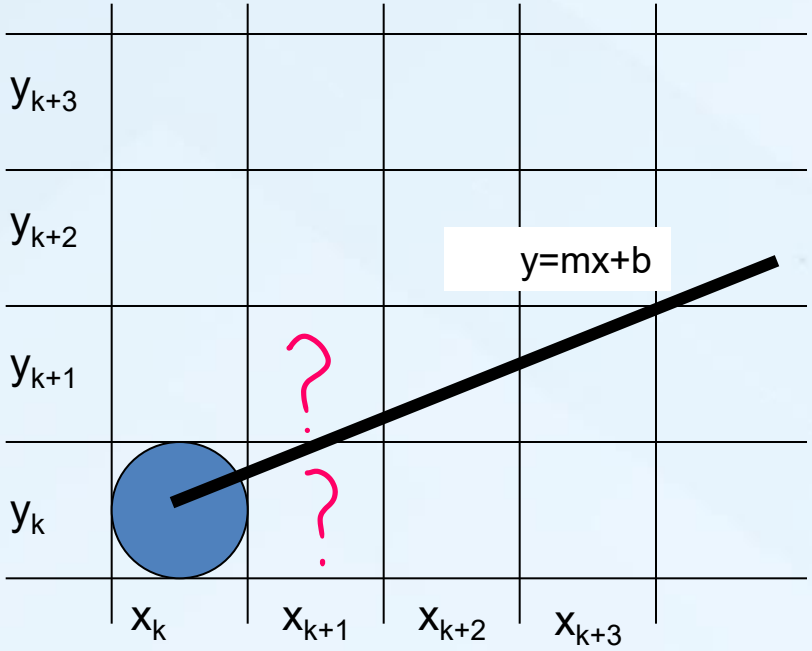
Bresenham's Line Algorithm



Decide what is the next pixel position (11,11) or (11,12)

Illustrating Bresenham's Approach





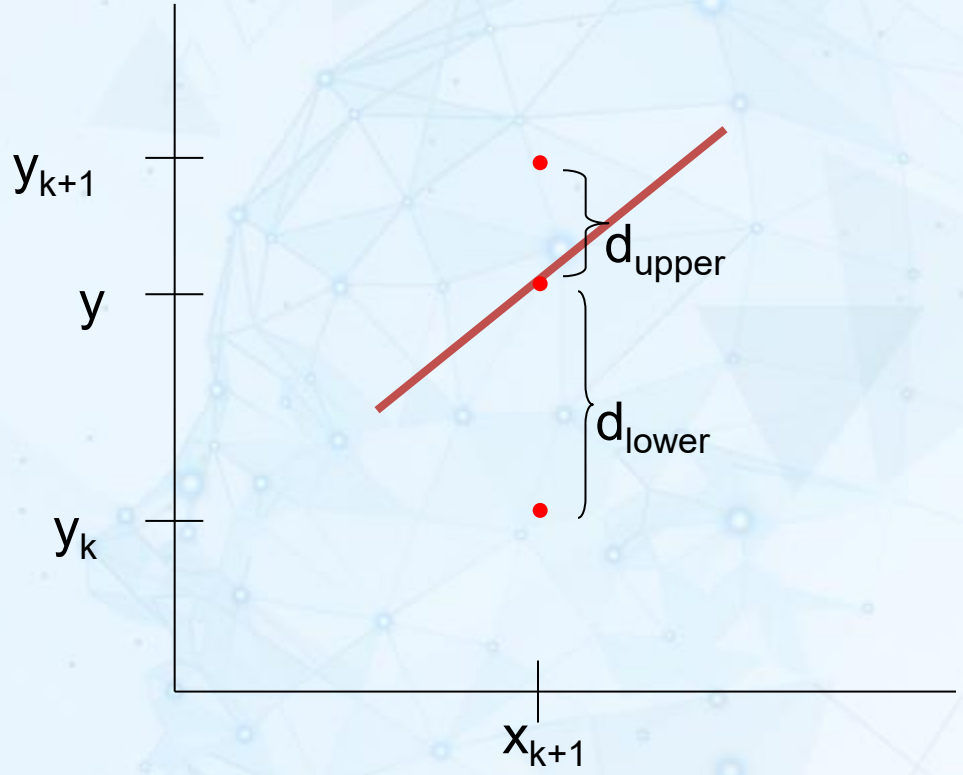
Because **m <1**
Then x++ (as the line will be more horizontally than vertically)

For the pixel position $x_{k+1}=x_k+1$, **which one we should choose?**
$(\mathbf{x_{k+1}},y_k)$ or $(\mathbf{x_{k+1}}, y_{k+1})$??

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Bresenham's Approach

- $y=m(x_k + 1)+b$
- $d_{lower}=y-y_k=m(x_k + 1)+b-y_k$
- $d_{upper}=(y_k+1)-y= y_k+1 -m(x_k + 1)-b$



- $d_{lower}- d_{upper}= 2m(x_k + 1)+2y_k+2b-1$
- Rearrange it to have integer calculations:

  $m=\Delta y/\Delta x$

  Decision parameter: $p_k= \Delta x(d_{lower}- d_{upper})=2\Delta y.x_k - 2\Delta x. y_k + c$

The Decision Parameter

Decision parameter: $p_k = \Delta x(d_{lower} - d_{upper}) = 2\Delta y.x_k - 2\Delta x.\ y_k + c$

- p_k has the same sign with $d_{lower} - d_{upper}$ since $\Delta x > 0$.
- c is constant and has the value $c = 2\Delta y + \Delta x(2b-1)$
  - c is independent of the pixel positions and is eliminated from decision parameter p_k.

- If $d_{lower} < d_{upper}$ then p_k is negative.
  - Plot the lower pixel (East)
- Otherwise
  - Plot the upper pixel (North East)

Succesive decision parameter

- At step k+1

  $p_{k+1}= 2\Delta y.x_{k+1} - 2\Delta x. y_{k+1} + c$

- Subtracting two subsequent decision parameters yields:

  $p_{k+1}-p_k= 2\Delta y.(x_{k+1}-x_k) - 2\Delta x. (y_{k+1}-y_k)$

- $x_{k+1}=x_k+1$ so

  $p_{k+1}= p_k + 2\Delta y - 2\Delta x. (y_{k+1}-y_k)$

  – $y_{k+1}-y_k$ is either 0 or 1 depending on the sign of p_k
- First parameter p_0

  – $p_0=2\ \Delta y - \Delta x$

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Bresenham's Line-Drawing Algorithm for I m I < 1

1. Input the two line endpoints and store the left endpoint in (x_0, y_0).

2. Load (x_0, y_0) into the frame buffer; that is, plot the first point.

3. Calculate constants Δx, Δy, $2\Delta y$, and $2\Delta y - 2\Delta x$, and obtain the starting value for the decision parameter as

$$p_0 = 2\,\Delta y - \Delta x$$

4. At each x_k along the line, starting at k = 0, perform the following test:

   If $p_k < 0$, the next point to plot is (x_{k+1}, y_k) and

$$p_{k+1} = p_k + 2\Delta y$$

   Otherwise, the next point to plot is (x_{k+1}, y_{k+1}) and

$$p_{k+1} = p_k + 2\Delta y - 2\Delta x$$

5. Repeat step 4 Δx -1 times.

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Bresenham's Line-Drawing Algorithm for I m I < 1

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

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Bresenham's Line-Drawing Algorithm for |m| < 1

Bresenham's Line drawing Algo. (3 Steps)

1. $P_0 = 2dy - dx$

each step

2. next point (x_{k+1}, y_k)

3. if $P_k < 0$ → $P_{k+1} = P_k + 2dy$

if $P_k \geq 0$ → next point (x_{k+1}, y_{k+1})

$P_{k+1} = P_k + 2dy - 2dx$

- Bres. Algo. use → Int numbers only
- high speed

→ use this Algo. to draw any line, any octant 1/8



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Trivial Situations: Do not need Bresenham

- $m = 0 \quad \Rightarrow \quad \text{horizontal line}$
- $m = \pm 1 \quad \Rightarrow \quad \text{line } y = \pm x$
- $m = \infty \quad \Rightarrow \quad \text{vertical line}$

Example

- Draw the line with endpoints (20,10) and (30, 18).
  - $\Delta x=30-20=10$, $\Delta y=18-10=8$,
  - $p_0 = 2\Delta y - \Delta x=16-10=6$
  - $2\Delta y=16$, and $2\Delta y - 2\Delta x=-4$
- Plot the initial position at (20,10), then

k	p_k	(x_{k+1}, y_{k+1})	k	p_k	(x_{k+1}, y_{k+1})
0	6	(21, 11)	5	6	(26, 15)
1	2	(22, 12)	6	2	(27, 16)
2	−2	(23, 12)	7	−2	(28, 16)
3	14	(24, 13)	8	14	(29, 17)
4	10	(25, 14)	9	10	(30, 18)

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For the line from **(20,10)** to **(30,18)**:

$$(x_1, y_1) = (20,10), \qquad (x_2, y_2) = (30,18)$$

1. Calculate Δx and Δy

$$\Delta x = x_2 - x_1 = 30 - 20 = 10$$

$$\Delta y = y_2 - y_1 = 18 - 10 = 8$$

Since:

$$0 < \frac{\Delta y}{\Delta x} < 1$$

we use the Bresenham case where x **always increases by 1.**

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

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The decision parameter is:

$$p_0 = 2\Delta y - \Delta x$$

Therefore:

$$p_0 = 2(8) - 10$$

$$p_0 = 6$$

We also calculate:

$$2\Delta y = 16$$

and

$$2\Delta y - 2\Delta x = 16 - 20 = -4$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

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So there are two possible updates:

$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

If $p_k < 0$:

Choose **E** pixel:

$$(x_{k+1}, y_{k+1}) = (x_k + 1, y_k)$$

and

$$p_{k+1} = p_k + 2\Delta y$$

Therefore:

$$p_{k+1} = p_k + 16$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

If $p_k \geq 0$:

Choose **NE** pixel:

$$(x_{k+1}, y_{k+1}) = (x_k + 1, y_k + 1)$$

and

$$p_{k+1} = p_k + 2\Delta y - 2\Delta x$$

Therefore:

$$p_{k+1} = p_k - 4$$

$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

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Step 0

Starting pixel:

$$(x_0, y_0) = (20, 10)$$

We already calculated:

$$p_0 = 6$$

Since:

$$p_0 = 6 \geq 0$$

we choose **NE**:

$$x_1 = 20 + 1 = 21$$

$$y_1 = 10 + 1 = 11$$

Now calculate p_1:

$$p_1 = p_0 + 2\Delta y - 2\Delta x$$

$$p_1 = 6 + 16 - 20$$

$$p_1 = 2$$

So:

$$(20, 10) \rightarrow (21, 11)$$

$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

Step 1

Current pixel:

$$(21, 11)$$

Current decision parameter:

$$p_1 = 2$$

Since:

$$2 \geq 0$$

choose NE.

New pixel:

$$x_2 = 21 + 1 = 22$$

$$y_2 = 11 + 1 = 12$$

$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

Calculate:

$$p_2 = p_1 + 2\Delta y - 2\Delta x$$

$$p_2 = 2 + 16 - 20$$

$$p_2 = -2$$

Therefore:

$$(21, 11) \rightarrow (22, 12)$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

Step 2

Current pixel:

$$(22, 12)$$

Current parameter:

$$p_2 = -2$$

Since:

$$-2 < 0$$

choose E.

So y does not increase:

$$x_3 = 22 + 1 = 23$$

$$y_3 = 12$$

Now calculate:

$$p_3 = p_2 + 2\Delta y$$

$$p_3 = -2 + 16$$

$$p_3 = 14$$

Therefore:

$$(22, 12) \rightarrow (23, 12)$$

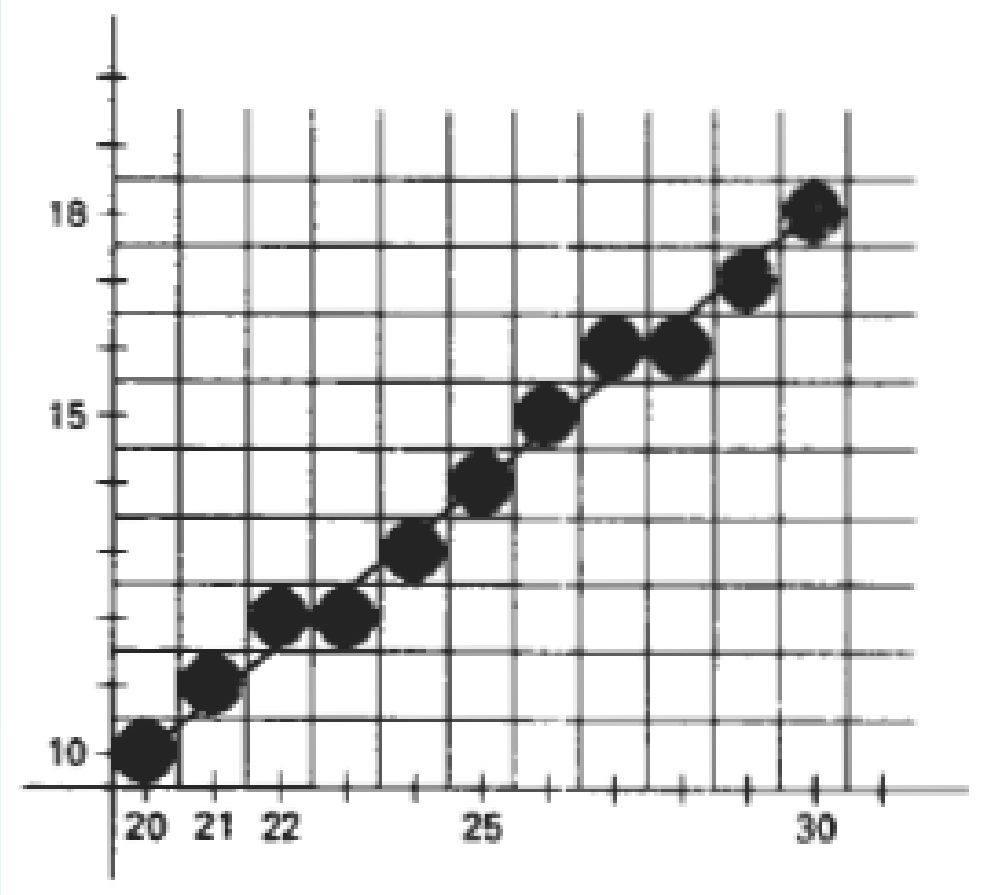
$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.

k	Current pixel (x_k, y_k)	p_k	Decision	Calculation of p_{k+1}	p_{k+1}	Next pixel
0	(20,10)	6	NE	$6 + 16 - 20$	2	(21,11)
1	(21,11)	2	NE	$2 + 16 - 20$	-2	(22,12)
2	(22,12)	-2	E	$-2 + 16$	14	(23,12)
3	(23,12)	14	NE	$14 + 16 - 20$	10	(24,13)
4	(24,13)	10	NE	$10 + 16 - 20$	6	(25,14)
5	(25,14)	6	NE	$6 + 16 - 20$	2	(26,15)
6	(26,15)	2	NE	$2 + 16 - 20$	-2	(27,16)
7	(27,16)	-2	E	$-2 + 16$	14	(28,16)
8	(28,16)	14	NE	$14 + 16 - 20$	10	(29,17)
9	(29,17)	10	NE	$10 + 16 - 20$	6	(30,18)

$$\begin{cases} p_k \geq 0 & \Rightarrow p_{k+1} = p_k - 4 \quad \text{(NE)} \\ p_k < 0 & \Rightarrow p_{k+1} = p_k + 16 \quad \text{(E)} \end{cases}$$

1. Find $\Delta x = x_2 - x_1$.
2. Find $\Delta y = y_2 - y_1$.
3. Calculate $p_0 = 2\Delta y - \Delta x$.
4. Start from (x_1, y_1).
5. Plot the current point.
6. If $p_k \geq 0$, increase **x and y** by 1.
7. Calculate $p_{k+1} = p_k + 2\Delta y - 2\Delta x$.
8. If $p_k < 0$, increase **x only** by 1.
9. Calculate $p_{k+1} = p_k + 2\Delta y$.
10. Repeat until the endpoint is reached.



Bresenham Line from (20,10) to (30,18)

k	P_k	(X_{k+1}, Y_{k+1})
0	6	(21, 11)
1	2	(22, 12)
2	-2	(23, 12)
3	14	(24, 13)
4	10	(25, 14)
5	6	(26, 15)
6	2	(27, 16)
7	-2	(28, 16)
8	14	(29, 17)
9	10	(30, 18)

Example

- Line end points:

$$(x_0, y_0) = (5,8); \quad (x_1, y_1) = (9,11)$$

- Deltas:

$$dx = 4; dy = 3$$

initially $p(5,8) = 2(dy)\text{-}(dx)$

$$= 6 - 4 = 2 > 0$$

$$p = 2 \quad \Rightarrow \quad NE$$

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- Line end points:

$$(x_0, y_0) = (20, 10): \qquad (x_1, y_1) = (30, 18)$$

- Deltas:

$$dx = 10, \qquad dy = 8$$

- Initially:

$$p(20, 10) = 2(dy) - (dx)$$

$$= 2(8) - 10 = 16 - 10 = 6 > 0$$

$$p = 6 \quad \Rightarrow \quad NE$$

- Next point:

$$(20, 10) \rightarrow (21, 11)$$

- Update p:

$$p = 6 + 2(dy) - 2(dx)$$

$$= 6 + 16 - 20 = 2 > 0$$

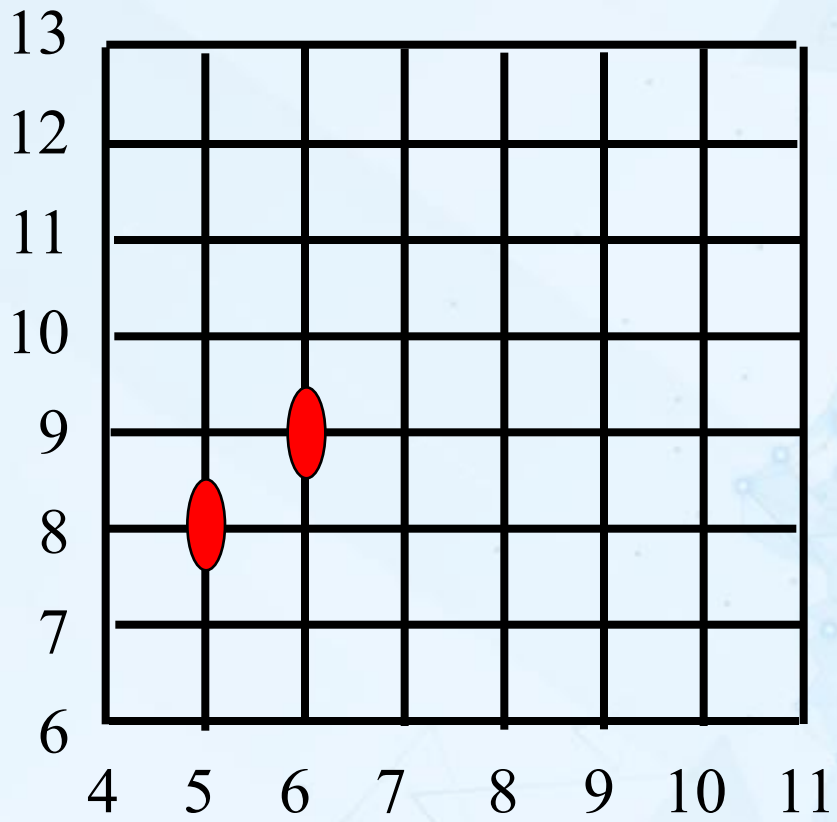
$$p = 2 \quad \Rightarrow \quad NE$$

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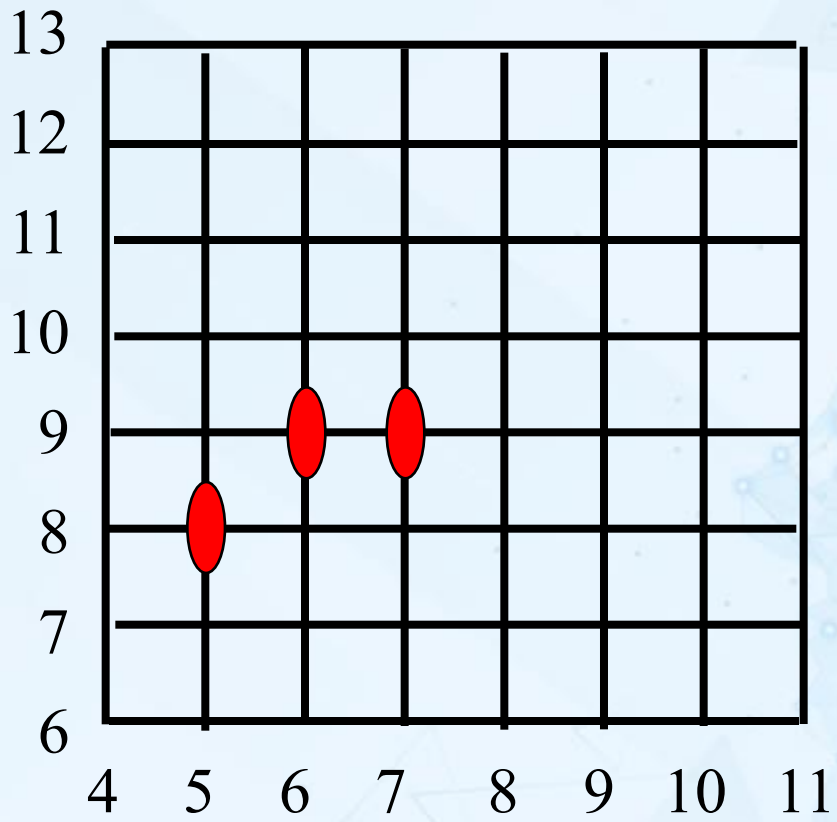
Then continue the same way:

Point	p	Direction
$(20, 10)$	6	NE
$(21, 11)$	2	NE
$(22, 12)$	-2	E
$(23, 12)$	14	NE
$(24, 13)$	10	NE
$(25, 14)$	6	NE
$(26, 15)$	2	NE
$(27, 16)$	-2	E
$(28, 16)$	14	NE
$(29, 17)$	10	NE
$(30, 18)$	$-$	$-$

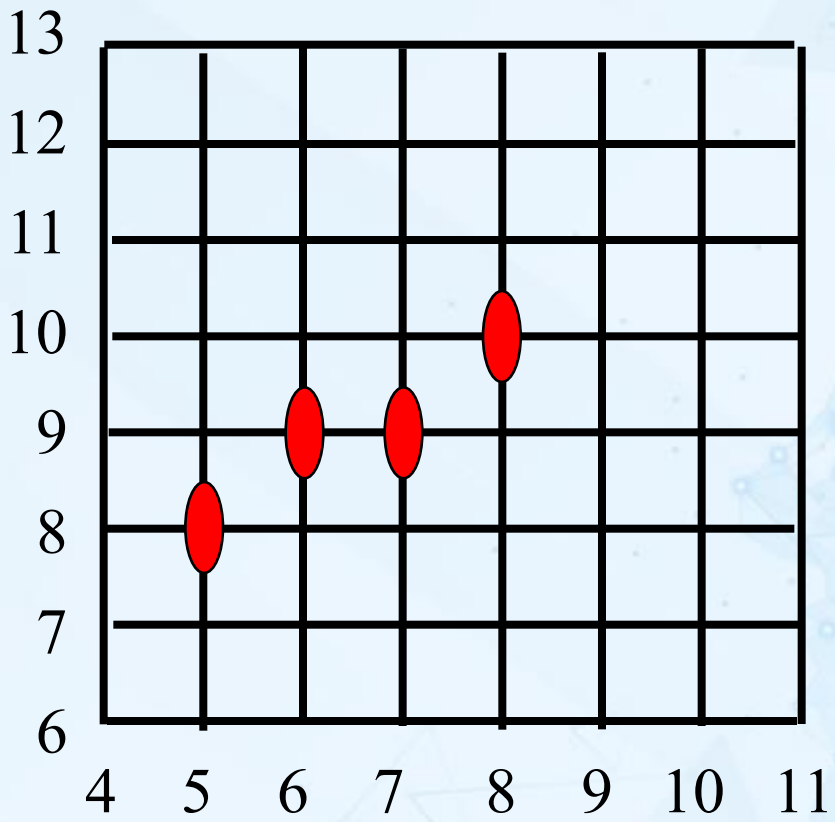
Graph



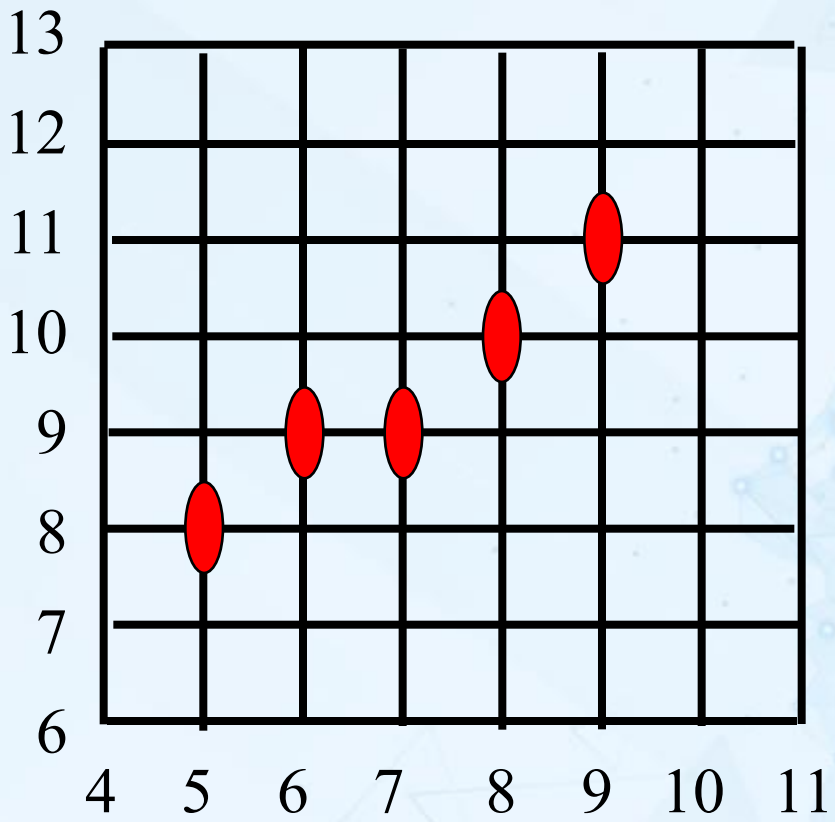
Continue the process...



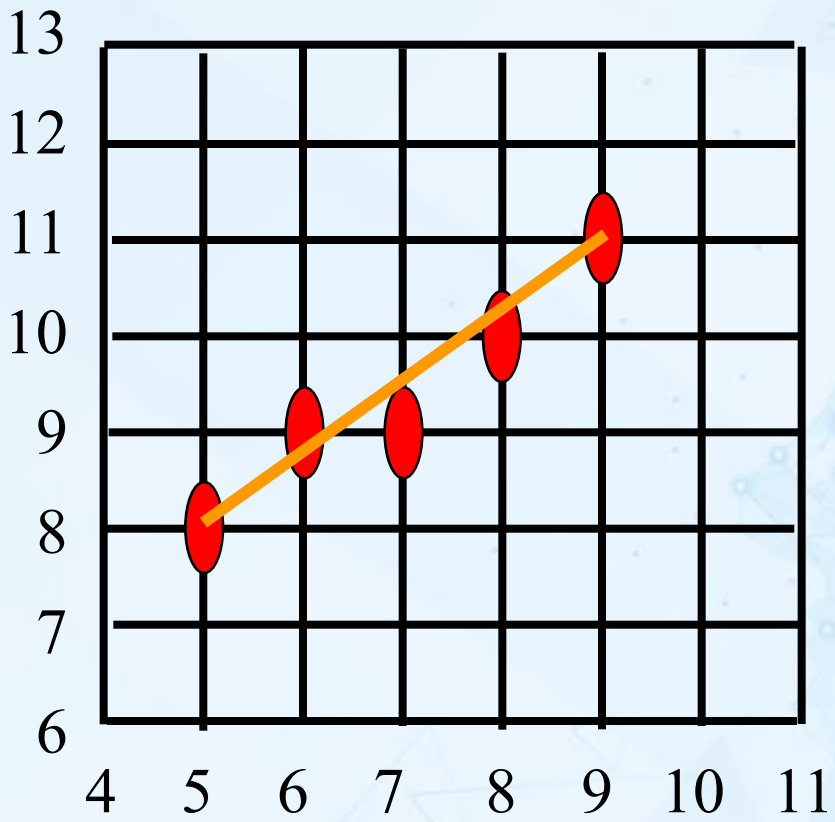
Graph

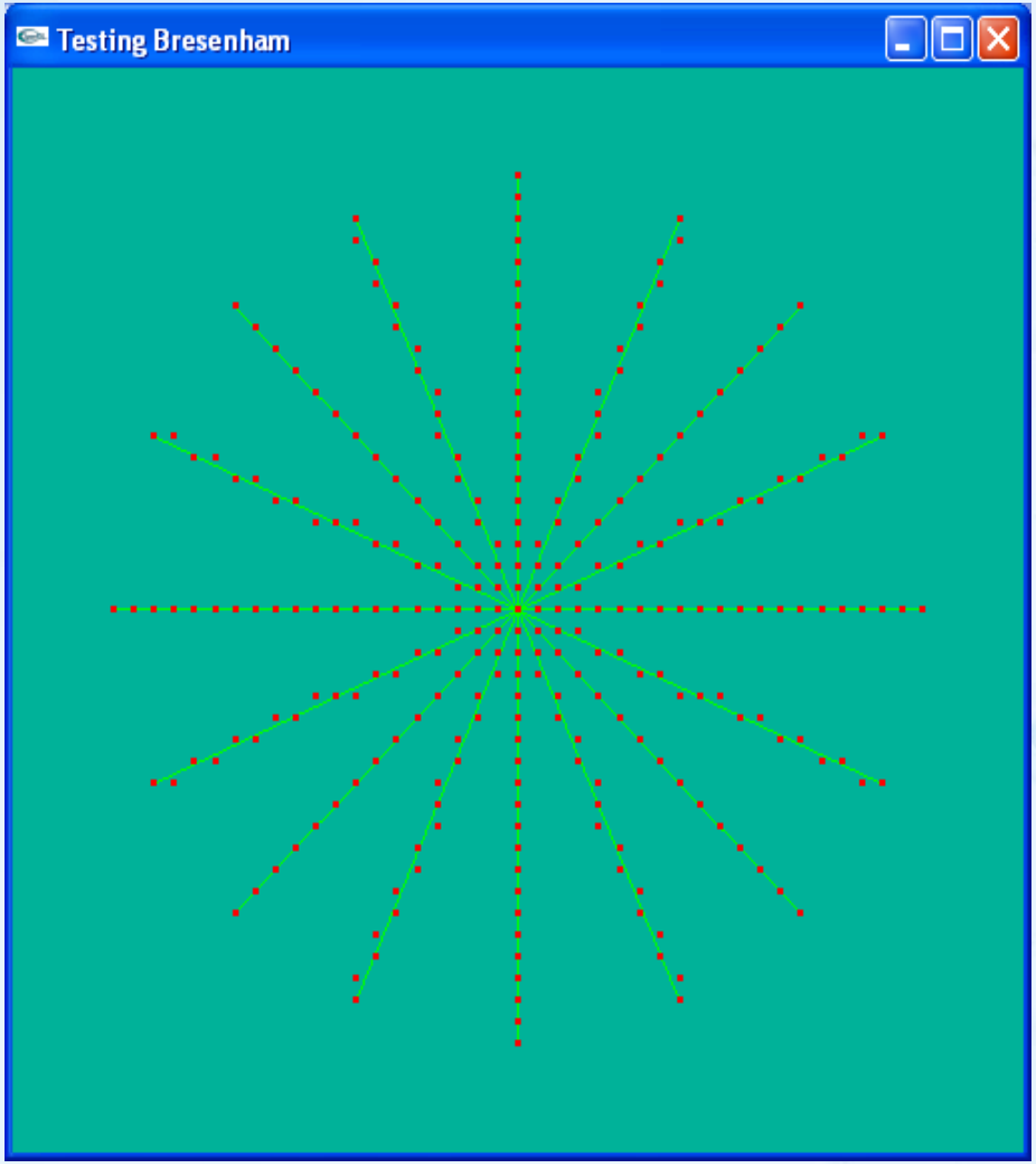


Graph



Graph



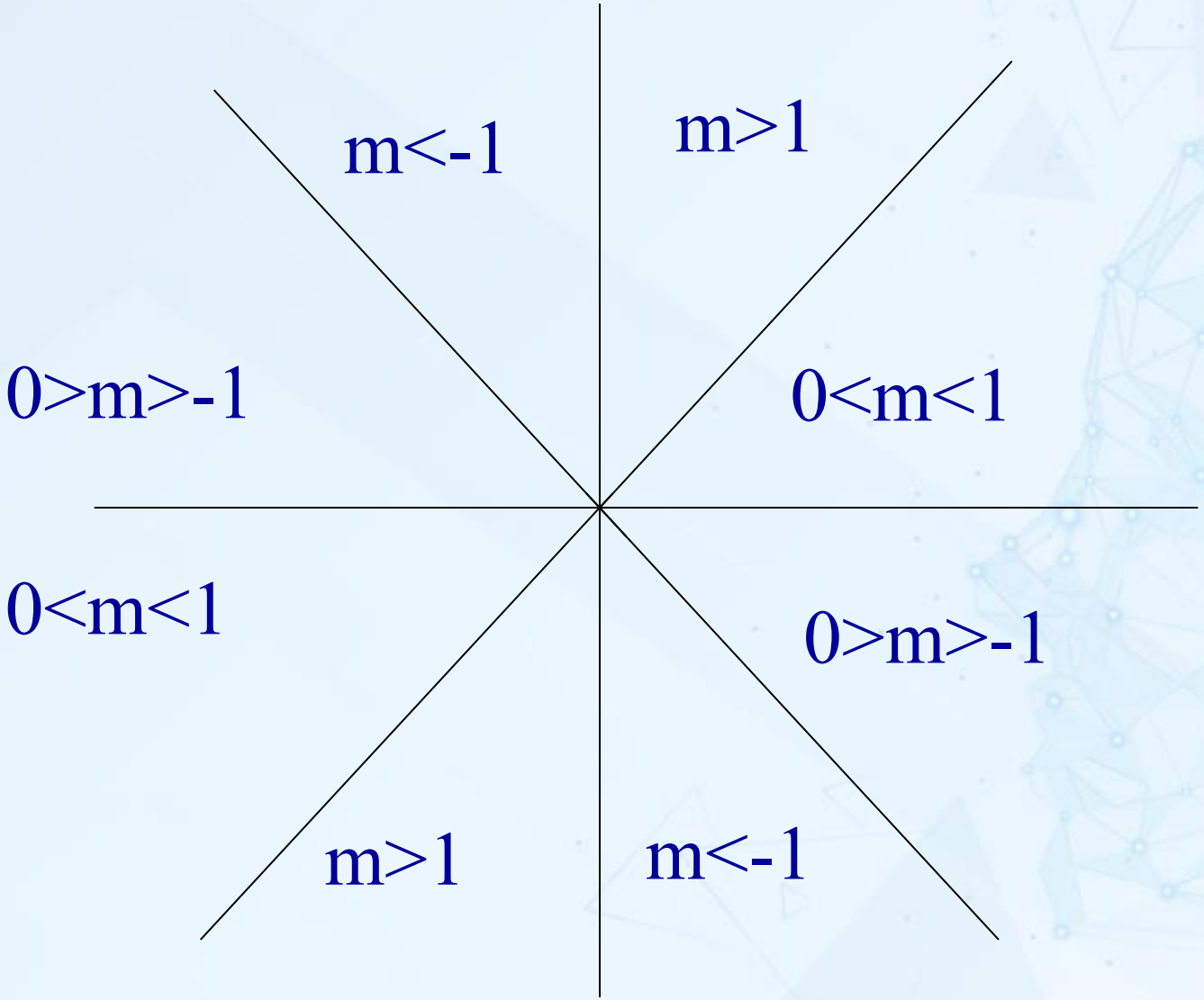


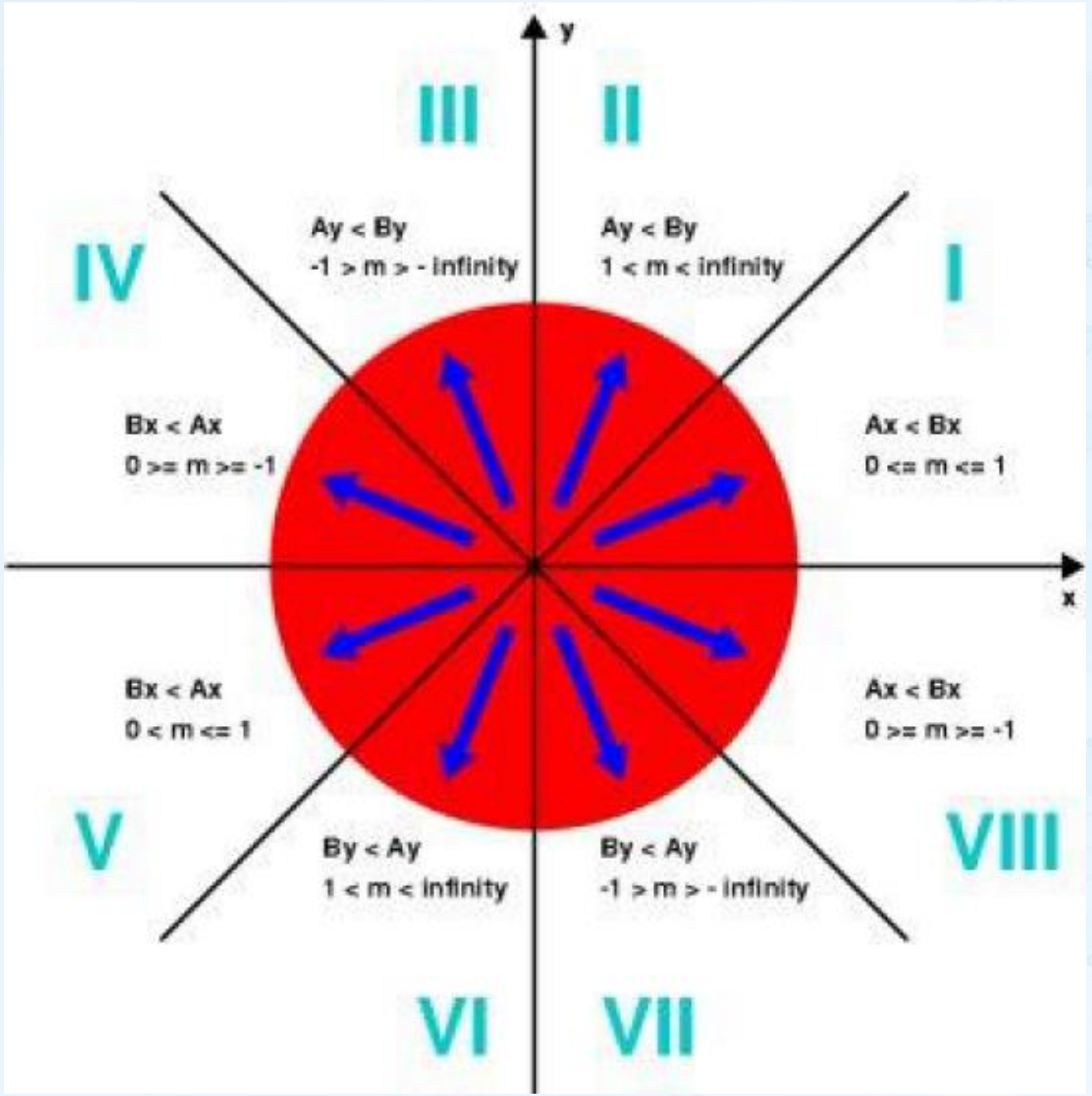
Simulating the Bresenham algorithm in drawing 8 radii on a circle of radius 20

Horizontal, vertical and ±45° radii handled as special cases

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Line-drawing algorithm should work in every octant, and special cases




y
x
III
II
IV
I
V
VIII
VI
VII
Ay < By
-1 > m > - infinity
Ay < By
1 < m < infinity
Bx < Ax
0 >= m >= -1
Ax < Bx
0 <= m <= 1
Bx < Ax
0 < m <= 1
Ax < Bx
0 >= m >= -1
By < Ay
1 < m < infinity
By < Ay
-1 > m > - infinity

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- each line for each octant ($\frac{1}{8}$) has characteristic

① First octant



* Ax < Bx
  - first x is less than 2nd X

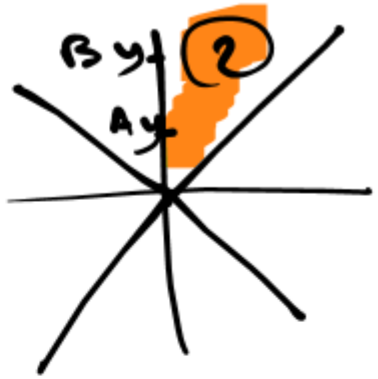
* Slope (m) $\Rightarrow$ $0 \leq m \leq 1$

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(2) 2nd octant

* 1st y < 2nd y

Ay < By



* Slope ⇒ 1 < m < Infinity

(3) Third octant

* Ay < By

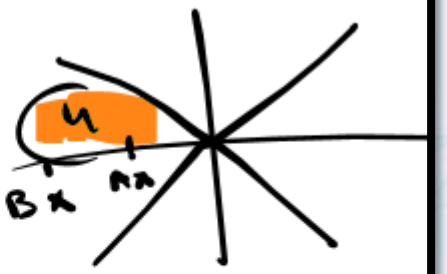
* $-1 > m > -\infty$



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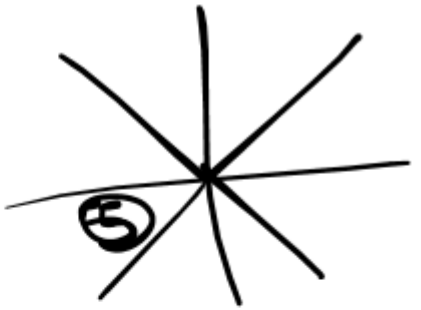
(4) Fourth octant

* $BX < AX$ (-ve)
* $0 < m \leq 1$



(5) Fifth octant

* $BX < AX$
* $0 < m \leq 1$



EELU

(6) octant

* $B_y < A_y$

* $1 < m < \infty$

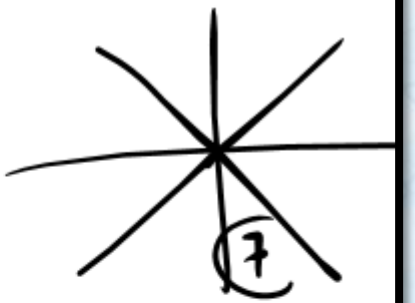


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(7) octant

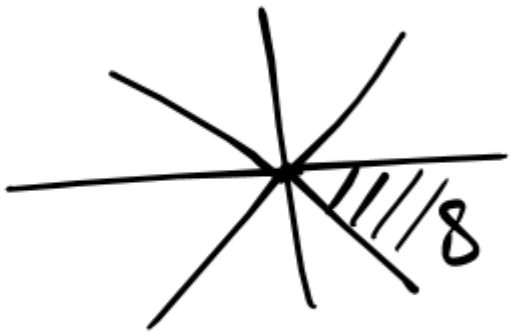
* $B_y < A_y$

* $-1 > m > -\infty$



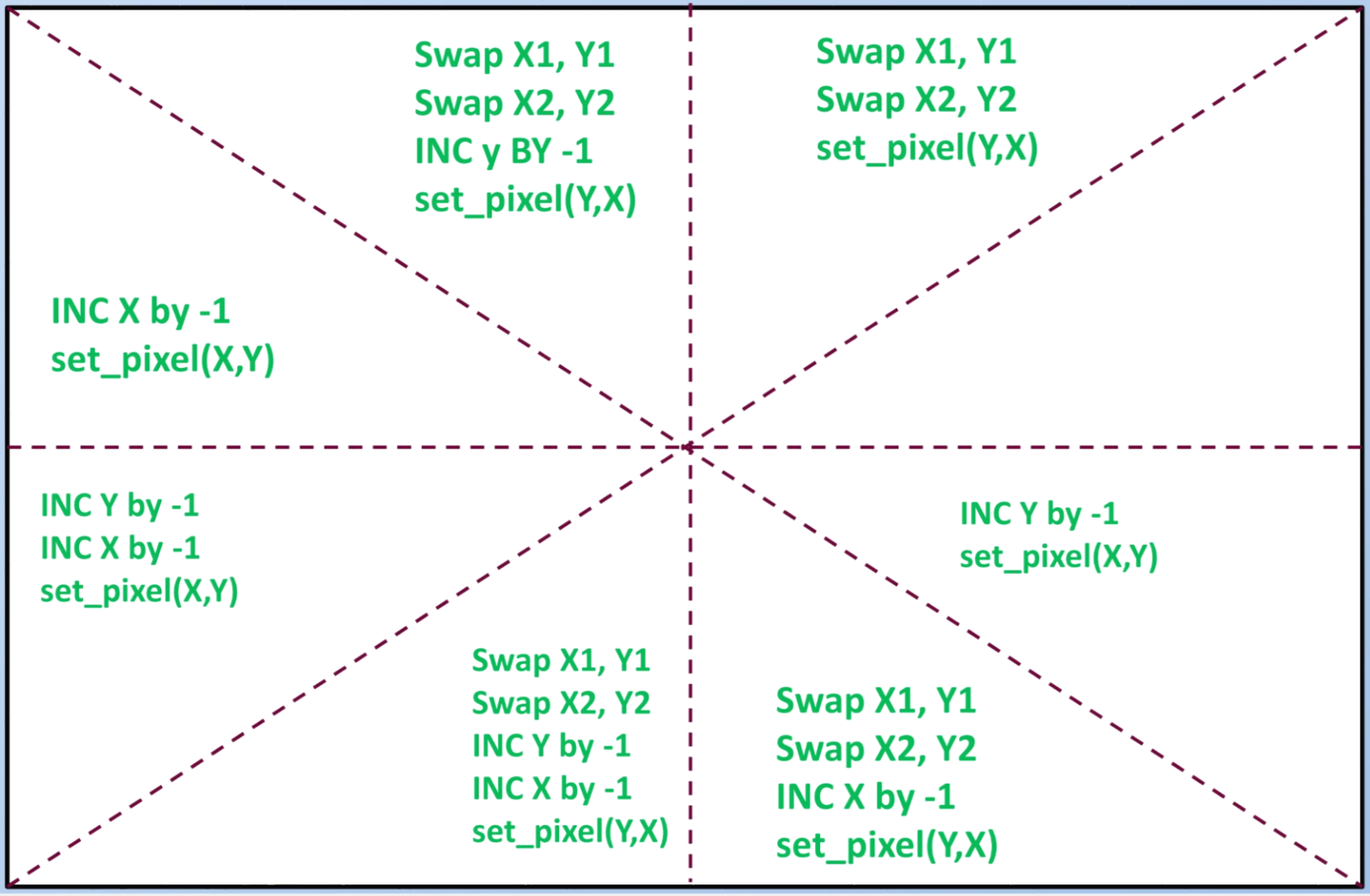
EELU

(8) octant



* $Ax < Bx$
* $0 \geq m \geq -1$

Bresenham's Algorithm



How to use Bresenham Algo. to draw any line in 8 octants

Two Rules

① cal. slope $m = \frac{\Delta y}{\Delta x}$

↳ if $|m| > 1$

↳ swap x values & y values

② cal. Δx & Δy

- if $\Delta x < 0$ or $\Delta y < 0$

↳ use $|\Delta x|$ & $|\Delta y|$

- but decrement x or y not increment

ex if $\Delta x < 0 \rightarrow x--$

if $\Delta y < 0 \rightarrow y--$

EELU

First Octant

First Point (32, 3) Second Point (48, 8)

$$\text{Slope} = \frac{8-3}{48-32} = 0.3125$$

Since x1 < x2 (32 < 48)

The line is in the first octant

dx = 16

dy = 5

2dy = 10

2dy-2dx = -22

P0 = 2dy-dx

t	P	X	Y
0	-6	32	3
1	4	33	3
2	-18	34	4
3	-8	35	4
4	2	36	4
5	-20	37	5
6	-10	38	5
7	0	39	5
8	-22	40	6
9	-12	41	6
10	-2	42	6
11	8	43	6
12	-14	44	7
13	-4	45	7
14	6	46	7
15	-16	47	8
16	-6	48	8

Since the slope is **less than 1**, we increase **X by 1 each step**.

EELU

$m > 1 \Rightarrow$ 2nd octant

And

$y_1 < y_2$

Second Octant

First Point (3, 32) Second Point (8, 48)

$$\text{Slope} = \frac{48-32}{8-3} = 3.2$$

Since y1 < y2 (32 < 48)

The line is in the second octant

- Switch x1 and y1 (32, 3)
- Switch x2 and y2 (48, 8)
- Apply Bressenham's algorithm
- Switch X and Y

dx = 16
dy = 5
2dy = 10
2dy-2dx = -22
P0 = 2dy-dx

t	P	X	Y
0	-6	32	3
1	4	33	3
2	-18	34	4
3	-8	35	4
4	2	36	4
5	-20	37	5
6	-10	38	5
7	0	39	5
8	-22	40	6
9	-12	41	6
10	-2	42	6
11	8	43	6
12	-14	44	7
13	-4	45	7
14	6	46	7
15	-16	47	8
16	-6	48	8

Y	X
3	32
3	33
4	34
4	35
4	36
5	37
5	38
5	39
6	40
6	41
6	42
6	43
7	44
7	45
7	46
8	47
8	48

Ex (3,32) (8,48) (2nd octant)

1) $m = \frac{\Delta y}{\Delta x} = \frac{48-32}{8-3} = 3.2$ ($|m| > 1$)

$\therefore |m| > 1 \longrightarrow$ swap x & y vals

↳ (32,3) (48,8)

→ use Bresenham steps

② Δx & Δy → +ve

$\Delta x > 0$ & $\Delta y > 0$

use Bresenham to get

x	y
32	3
33	4

③ Reswap x & y vals

x	y
3	32
4	33

EELU

Third Octant

First Point (8, 32) Second Point (3, 48)

$$\text{Slope} = \frac{48-32}{3-8} = -3.2$$

Since y1 < y2 (32 < 48)

The line is in the Third octant

- Swap x1 and y1 (32, 8)
- Swap x2 and y2 (48, 3)
- Set dy = -dy
- Decrement Y
- Switch X and Y

dx = 16
dy = -5 → dy = 5
2dy = 10
2dy-2dx = -22
P0 = 2dy - dx

t	P	X	Y
0	-6	32	8
1	4	33	8
2	-18	34	7
3	-8	35	7
4	2	36	7
5	-20	37	6
6	-10	38	6
7	0	39	6
8	-22	40	5
9	-12	41	5
10	-2	42	5
11	8	43	5
12	-14	44	4
13	-4	45	4
14	6	46	4
15	-16	47	3
16	-6	48	3

Y	X
8	32
8	33
7	34
7	35
7	36
6	37
6	38
6	39
5	40
5	41
5	42
5	43
4	44
4	45
4	46
3	47
3	48

Fourth Octant

First Point (48, 3) Second Point (32, 8)

$$\text{Slope} = \frac{8-3}{32-48} = -0.3125$$

Since x1 > x2 (48 > 32)

The line is in the Fourth octant

- Set dx = -dx
- Decrement x

- dx = -16 → dx = 16
- dy = 5
- 2dy = 10
- 2dy-2dx = -22
- P0 = 2dy-dx

t	P	X	Y
0	-6	48	3
1	4	47	3
2	-18	46	4
3	-8	45	4
4	2	44	4
5	-20	43	5
6	-10	42	5
7	0	41	5
8	-22	40	6
9	-12	39	6
10	-2	38	6
11	8	37	6
12	-14	36	7
13	-4	35	7
14	6	34	7
15	-16	33	8
16	-6	32	8

EELU

Fifth Octant

First Point (48, 8) Second Point (32, 3)

$$\text{Slope} = \frac{3-8}{32-48} = 0.3125$$

Since x1 > x2 (48 > 32)

The line is in the Fifth octant

- Set dx = -dx
- Set dy = -dy
- Decrement x
- Decrement y

- dx = -16 → dx = 16
- dy = -5 → dy = 5
- 2dy = 10
- 2dy-2dx = -22
- P0 = 2dy - dx

t	P	X	Y
0	-6	48	8
1	4	47	8
2	-18	46	7
3	-8	45	7
4	2	44	7
5	-20	43	6
6	-10	42	6
7	0	41	6
8	-22	40	5
9	-12	39	5
10	-2	38	5
11	8	37	5
12	-14	36	4
13	-4	35	4
14	6	34	4
15	-16	33	3
16	-6	32	3

EELU

Sixth Octant

First Point (8, 48)

Second Point (3, 32)

$$\text{Slope} = \frac{32-48}{3-8} = 3.2$$

Since y1 > y2 (48 > 32)

The line is in the sixth octant

- Swap x1 and y1 (48, 8)
- Swap x2 and y2 (32, 3)
- Set dx = -dx
- Set dy = -dy
- Decrement x
- Decrement y
- Swap X and Y

dx = -16 → dx = 16

dy = -5 → dy = 5

2dy = 10

2dy-2dx = -22

P0 = 2dy-dx

t	P	X	Y
0	-6	48	8
1	4	47	8
2	-18	46	7
3	-8	45	7
4	2	44	7
5	-20	43	6
6	-10	42	6
7	0	41	6
8	-22	40	5
9	-12	39	5
10	-2	38	5
11	8	37	5
12	-14	36	4
13	-4	35	4
14	6	34	4
15	-16	33	3
16	-6	32	3

Y	X
8	48
8	47
7	46
7	45
7	44
6	43
6	42
6	41
5	40
5	39
5	38
5	37
4	36
4	35
4	34
3	33
3	32

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Seventh Octant

First Point (3, 48)

Second Point (8, 32)

$$\text{Slope} = \frac{32-48}{8-3} = -3.2$$

Since y1 > y2 (48 > 32)

The line is in the seventh octant

- Swap x1 and y1 (48, 3)
- Swap x2 and y2 (32, 8)
- Set dx = -dx
- Decrement x
- Swap X and Y

dx = -16 → dx = 16
dy = 5
2dy = 10
2dy-2dx = -22
P0 = 2dy-dx

t	P	X	Y
0	-6	48	3
1	4	47	3
2	-18	46	4
3	-8	45	4
4	2	44	4
5	-20	43	5
6	-10	42	5
7	0	41	5
8	-22	40	6
9	-12	39	6
10	-2	38	6
11	8	37	6
12	-14	36	7
13	-4	35	7
14	6	34	7
15	-16	33	8
16	-6	32	8

Y	X
3	48
3	47
4	46
4	45
4	44
5	43
5	42
5	41
6	40
6	39
6	38
6	37
7	36
7	35
7	34
8	33
8	32

EELU

Eighth Octant

First Point (32, 8) Second Point (48, 3)

$$\text{Slope} = \frac{3-8}{48-32} = -0.3125$$

Since x1 < x2 (32 < 48)

The line is in the Eighth octant

- Set dy = -dy
- Decrement y

- dx = 16
- dy = -5 → dy = 5
- 2dy = 10
- 2dy-2dx =-22
- P0 = 2dy-dx

t	P	X	Y
0	-6	32	8
1	4	33	8
2	-18	34	7
3	-8	35	7
4	2	36	7
5	-20	37	6
6	-10	38	6
7	0	39	6
8	-22	40	5
9	-12	41	5
10	-2	42	5
11	8	43	5
12	-14	44	4
13	-4	45	4
14	6	46	4
15	-16	47	3
16	-6	48	3

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THANK YOU FOR WATCHING

QUESTIONS?

